

RAMPRASAD SUBRAMANIAN

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SUMMARY

M.Sc. in Data Science with a B.Sc. in Applied Mathematics, seeking entry-level data scientist or data analyst roles. Strongest in experimentation, statistical inference, predictive modeling, forecasting and model deployment.

EDUCATION

University of Europe for Applied Sciences | Potsdam, Germany | Sept. 2024 – Aug. 2026

Master of Science in Data Science

FLAME University | Pune, India | May 2023

Bachelor of Science in Applied Mathematics

PROJECTS

Telco Churn Risk API | *Python, scikit-learn, SHAP, FastAPI, Docker, GitHub Actions*

- Compared three models on ~7,000 telecom customers with 5-fold CV; logistic regression performed best (ROC-AUC 0.845) and was tuned and calibrated
- Set a cost-based decision threshold from assumed retention costs, for an estimated \$106,930 saved per 1,000 customers scored versus flagging no one
- Deployed as a Dockerized FastAPI service with SHAP explanations (p95 latency 17 ms, 94% test coverage, CI)

Deutsche Bahn Punctuality Analysis | *SQL, dbt, DuckDB, Streamlit, GitHub Actions*

- Built a tested analytics pipeline (Python ingestion, DuckDB, dbt staging/intermediate/mart models, Streamlit dashboard) on 491,562 long-distance stop events, with 45 dbt data tests and a monthly automated refresh
- Found trains spend 8% of scheduled time at stations but add 52% of net delay there (Aug. 2026; June: 51%)

Mobile Game A/B Test Analysis (Cookie Cats) | *Python, Jupyter*

- Analyzed a ~90,000-player test of paywall gate placement with power analysis, two-proportion z-tests, bootstrap CIs and Benjamini-Hochberg correction
- Recommended keeping the gate at level 30 (significantly higher day-7 retention, corrected $p = 0.0047$); simulated how peeking at six interim checks inflates false positives to 20%

M.Sc. Thesis: LLM-Guided Socratic Hint Generation | *PyTorch, Hugging Face, Unsloth, QLoRA* | 2026

- Fine-tuned Qwen2.5-3B-Instruct with QLoRA (6 GB VRAM) on MathDial tutoring dialogues to generate Socratic hints conditioned on an inferred student-mastery estimate
- Evaluated five conditions on 241 held-out dialogues: fine-tuning improved LLM-judged hint quality over the prompted base model, while student-state conditioning gave no significant gain across three seeds

CAISO Day-Ahead Electricity Demand Forecast | *Python, pandas, scikit-learn, Prophet, REST APIs*

- Built a pipeline pulling 2 years of hourly CAISO demand (EIA API) and temperature (Open-Meteo API), 17,544 rows, refreshed on each run
- Walk-forward backtest (61 windows): gradient boosting reached 2.89% MAPE (seasonal-naive baseline: 8.04%)

More on GitHub: aspect sentiment analysis (1.31M reviews), staggered difference-in-differences, CATE meta-learners, LaLonde causal benchmark, student-dropout prediction

TECHNICAL SKILLS

Programming: Python, R, SQL

Statistics: A/B testing, hypothesis testing, causal inference (difference-in-differences, propensity scores)

Machine Learning: classification, regression, time-series forecasting, feature engineering, calibration, SHAP

NLP & LLMs: transformer fine-tuning (BERT, RoBERTa), QLoRA, LLM evaluation

Libraries: pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, Prophet, tidyverse, dbt, DuckDB

Tools: FastAPI, Docker, Git, GitHub Actions (CI/CD), Streamlit, Tableau

EXPERIENCE

Corizo | Remote | June 2022 – Aug. 2022

Data Science Intern

- Built and evaluated predictive models in Python, including a wine-cultivar classifier

CERTIFICATIONS & LANGUAGES

Certification: AWS Academy Graduate – Cloud Foundations (AWS Academy, June 2025)

Languages: English, Tamil, Hindi